

	Internal Exam	Assignment	Project Evaluation	End Semester Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA2MN104 (P)
Course Title	Regression using JASP software
Type of Course	Minor
Semester	II

Academic Level	100 - 199				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Basic understanding of statistical concepts, familiarity with algebraic concepts				
Course Summary	Covering advanced statistical concepts like skewness, kurtosis, multiple regression, and JASP software utilization, alongside developing skills in descriptive statistics, data manipulation, result interpretation, and understanding sampling distributions and test statistics using JASP.				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Grasp skewness and kurtosis measures, emphasizing Pearson's measure and percentile measure and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Quiz
CO2	Understand multiple regression concepts and the selection process and analyze data to help entrepreneurial decisions using critical thinking skills..	U	F	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Proficiently use JASP software for statistical analysis and result interpretation.	U	C	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Comprehend sampling distributions and test statistics for Chi-square, F, and t distributions.	U	C	Instructor-created exams / Home Assignments
CO5	Implement theoretical knowledge in practical scenarios through hands-on exercises using JASP	Ap	C	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Illustrate how to draw scatter plot for correlation between variables.	Ap	P	Viva Voce/ Instructor-created exams
* - Remember (R), Understand (U), Apply (Ap), Analyse (An), Evaluate (E), Create (C)				

- Factual Knowledge(F) Conceptual Knowledge (C) Procedural Knowledge (P) Metacognitive Knowledge (M)
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Detailed Syllabus:

Module	Unit	Content	Hrs (45 +30)	Marks (70)
I	Skewness and Kurtosis		8	10
	1	Skewness	2	
	2	Kurtosis	2	
	3	Pearson's measure of skewness	2	
	4	Percentile measure of Kurtosis	2	
	Unit 1:3.13 Ref[2] Unit 2: 3.14 Ref[2] Unit 3: 3.13 Ref[2] Unit 4: 3.14 Ref[2]			
II	Multiple regression		12	25
	5	Multiple regression	1	
	6	Multiple Regression and Correlation Analysis	1	
	7	Assumptions of Linear Multiple Regression Analysis	1	
	8	Coefficient of Multiple Determination	1	
	9	Partial correlation	1	
	10	Partial correlation coefficient	2	
	11	The Significance of a Partial Correlation Coefficient	1	
	12	Multiple correlation	1	
	13	Coefficient of Multiple Correlation	1	
	14	Advantages of Multiple Correlation Analysis	1	

	15	Limitations of Multiple Correlation Analysis	1	
	Unit 5: 9.9 Ref[1] Unit 6: 9.9 Ref[1] Unit 7: 9.10 Ref[1] Unit 8: 9.7 Ref[1] Unit 9: 9.2 Ref[1] Unit 10: 9.2 Ref[1] Unit 11: 9.6 Ref[1] Unit 12: 9.7Ref[1] Unit 13: 9.7 Ref[1] Unit 14: 9.8 Ref[1] Unit 15: 9.9 Ref[1]			
III	JASPstatistical software		13	20
	16	Installing JASP	2	
	17	Loading data in JASP	2	
	18	Changing data from one measurement scale to another	3	
	19	Calculating Mean, Median and Mode in JASP	3	
	20	Calculating Range, standard deviation and variance using JASP	3	
	Unit 16: 3.1 Ref[3] Unit 17: 3.3 Ref[3] Unit 18: 3.5 Ref[3] Unit 19: 4.1 Ref[3] Unit 20: 4.2 Ref[3]			
IV	Sampling distributions		12	15
	21	Chi-square distribution	4	
	22	Student's t distribution	4	

	23	F distribution	4	
	Unit 21: 13.1 Ref[2] Unit 22: 14.2 Ref[2] Unit 23: 14.5 Ref[2]			
V	PRACTICUM		30	
	Do practice problems in JASP software from any 5 units of the given list and one additional problem decided by the teacher-in-charge, related to the content of the course. Other units listed here may be used as demonstrations of the concepts taught in the course.			
	1	Problems on plotting scatter plots		
	2	Correlation calculation		
	3	Interpretation of correlation coefficient in JASP		
	4	Finding Rank correlation		
	5	Introduce correlation matrix in JASP		
	6	Linear regression model		
	7	Model checking		
	8	Model selection		
	Books and References: Unit 1: 11.1.1 Ref[3] Unit 2: 11.1.3 Ref[3] Unit 3:11.1.5 Ref[3] Unit 4: 11.1.6 Ref[3] Unit 5: 11.2 Ref[3] Unit 6: 11.3 Ref[3] Unit 7:11.10 Ref[3] Unit 8:11.11 Ref[3]			
Books and References: 1. S.P Gupta (2021), Statistical Methods 46 th Edition 2. Gupta, S.C. and Kapoor, V.K. (1997) Fundamentals of Mathematical Statistics. Sultan Chand and Sons, New Delhi 3. Navarro, D.J., Foxcroft, D.R., & Faulkenberry, T.J. (2019). Learning Statistics with JASP: A Tutorial for Psychology Students and Other Beginners. (Version 1/(√2)).				

Mapping of COs with PSOs and POs :

	PSO 1	PSO 2	PSO 3	PSO4	PSO 5	PSO6	PO1	PO2	PO3	PO4	PO5	PO6
CO 1	2	1	-	-	-	-	1	2	-	-	-	3
CO 2	-	-	2	-	-	2	2	2	-	-	3	-
CO 3	3	2	3	-	2	3	3	3	2	3	-	-
CO 4	3	3	-	-	-	3	3	2	1	-	-	-
CO 5	-	-	3	2	-	-	2	1	-	3	-	-
CO 6	2	1	-	-	-	2	2	2	-	-	-	-

Correlation Levels:

Level	Correlation
-	Nil
1	Slightly / Low
2	Moderate / Medium
3	Substantial / High

Assessment Rubrics:

- Quiz / Assignment/ Quiz/ Discussion / Seminar
- Midterm Exam
- Programming Assignments (20%)
- Final Exam (70%)

Mapping of COs to Assessment Rubrics :

	Internal	Assignm	Project	End Semester
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	Exam	ent	Evaluation	Examinations
CO 1	✓	✓		✓
CO 2	✓	✓		✓
CO 3	✓			✓
CO 4		✓		✓
CO 5		✓		✓
CO 6	✓			

Programme	BSc Statistics
Course Code	STA3MN204 (P)
Course Title	Tests of hypothesis and SVM

Type of Course	Minor				
Semester	III				
Academic Level	200 - 299				
Course Details	Credit	Lecture per week	Tutorial per week	Practical per week	Total Hours
	4	3	-	2	75
Pre-requisites	Familiarity with algebraic concepts, basic statistics and probability concepts. Understanding of data visualization methods.				
Course Summary	Explore hypothesis testing basics like null and alternative hypotheses, critical regions, significance levels, and one/two-tailed tests, alongside t-tests, chi-square tests, and support vector machines, emphasizing practical applications with R				

Course Outcomes (CO):

CO	CO Statement	Cognitive Level*	Knowledge Category#	Evaluation Tools used
CO1	Grasp fundamentals of hypothesis testing, including null and alternative hypotheses, critical regions, and significance levels.	U	C	Instructor-created exams / Quiz
CO2	Understand small and large sample tests, focusing on tests for single mean and single proportion, t-tests, and chi-square tests.	U	C	Practical Assignment / Observation of Practical Skills/ Instructor-created exams
CO3	Interpret results and understand the implications of large and small sample tests in real life situations and analyze data to help entrepreneurial decisions using critical thinking skills.	U	F	Seminar Presentation / Group Tutorial Work/ Instructor-created exams
CO4	Explain the fundamental purpose of one way ANOVA and its purpose in statistical analysis and critically evaluate ethical implications of statistical methods aligning with human values.	U	C	Instructor-created exams / Home Assignments
CO5	Gain an overview of support vector machines, hyperplanes, and classifiers.	U	C	One Minute Reflection Writing assignments/ Instructor-created exams
CO6	Implement theoretical knowledge to practical scenarios through hands-on	Ap	P	Viva Voce/ Instructor-created